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MILITARY SPECIFICATION

FUSE, FIRECRACKER, 30-SECOND

1. SCOPE

1.1 Scope. - This specification covers one type of 30-second firecracker fuse.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

JAN-P-223 - Powder, Black

STANDARDS

MIL-STD-129 - Marking of Shipments

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 - General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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CODE OF FEDERAL REGULATIONS

49CFR 71 - 78 - Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition - Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Materials. - The materials used in the manufacture of the firecracker fuse shall be in accordance with this specification. Materials not specifically designated herein shall be of the best quality used for the purpose in commercial practice. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.2 Construction. - The 30-second firecracker fuse shall consist of a train of fine-grained black powder, complying with the requirements of Specification JAN-P-223, covered with two center cotton strands spun clockwise around it. This shall be covered with 12 intermediate binder strands uniformly and tightly twisted counterclockwise. In turn, this shall then be covered with six outer jacket strands twisted clockwise over the binder strands and waterproofed in accordance with 3.4.

3.3 Waterproofing. - The fuse shall be waterproofed by applying at least two coats of lacquer to the exterior surface. The first coat shall be well dried before applying the second coat. The waterproofing material shall be applied to the fuse in such a manner as to insure compliance with burning time requirements specified in 3.5.

3.4 Dimensions. - The outside diameter of the fuse shall be 0.10 ± 0.01 inch.

3.5 Burning time. - The burning time of the fuse shall be not less than 25 seconds nor more than 35 seconds per foot, when tested as specified in 4.5.

3.6 Workmanship. - Workmanship shall be of the highest grade throughout and in accordance with the best standard practice for this type of material. The finished fuse material shall be free of kinks, breaks, loose strands, and other evidence of non-conformity.

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4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 General. - The firecracker fuse covered by this specification shall be subject to all the tests as specified herein to determine compliance with the specified requirements. Testing shall be performed in the specified manner, but the Government reserves the right to specify any additional tests, that it may deem necessary, to determine compliance with the requirements.

4.1.1 Place of inspection and test. - Unless otherwise specified, final inspection for lot acceptance shall be made at the place of manufacture. The contractor shall furnish all necessary facilities and equipment for making tests and inspections required by this specification, and shall execute all tests under the supervision of the Government inspector.

4.1.2 Destroyed specimens. - All specimens destroyed in making tests required by this specification shall be in addition to the quantity specified in the contract or order and shall be furnished without increasing the cost of the contract or order.

4.2 Sampling. - The Government inspector shall select three representative pieces of the fuse, 3 feet in length, from each lot submitted for acceptance. A 12-inch piece from each of the three samples shall be used in the burning time test as specified in 4.5.

4.3 Inspection lot. - A lot shall consist of firecracker fuse manufactured under essentially the same conditions, and submitted for inspection at the same time.

4.3.1 Production lot size. - A lot shall consist of 5,000 feet of fuse.

4.4 Inspection. - The general inspection procedures shall be in accordance with the General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment, OP 400; Government inspection shall be in accordance with the provisions of General Procedures for Government Inspection, and Contractor's inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.4.1 Visual and dimensional inspection. - The sample pieces of fuse shall be visually and dimensionally inspected by the Government inspector to determine compliance with 3.4 and 3.7.

4.5 Burning time test. - Each of the three 12-inch pieces selected in accordance with 4.2 shall be tested for burning time by igniting one end and measuring the elapsed time from

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ignition to complete burn out at the opposite end to determine compliance with 3.5. A stop watch accurate to 1/10 of a second shall be used for measuring the elapsed time.

4.6 Retest and rejection

4.6.1 Retest. - When a lot fails the test as specified in 4.5, it shall be considered for retest under the following provisions:

(a) If not more than one 12-inch length of fuse fails to burn its complete length.

(b) If not more than two 12-inch lengths of fuse fail to comply with the burning time requirements.

For retest, six pieces 4 feet in length shall be selected and prepared as specified in 4.2 and double quantities of 12-inch pieces shall be subjected to the test specified in 4.5. For acceptance after retest, no failure of (a) above, and not more than one failure of (b) shall be permitted. Only one retest will be allowed.

4.6.2 Rejection. - A lot shall be rejected without retest for either of the following conditions.

(a) If one 12-inch length of fuse fails to burn its complete length together with one 12-inch length of fuse failing to comply with the burning time requirements.

(b) If more than one 12-inch length of fuse fails to burn its complete length.

(c) If all three 12-inch lengths tested fail to comply with the burning time requirements.

5. PREPARATION FOR DELIVERY

5.1 Precautions in storage. - After manufacture, and prior to shipment, the fuse shall be stored in dry, heated storage.

5.2 Packaging and packing. - Unless otherwise specified packaging and packing of the fuse shall be in accordance with the Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, Section 73.100 (o) (Safety fuse). Each container shall be waterproof, and contain 5,000 feet of fuse unless otherwise specified in the contract or order.

5.3 Marking

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5.3.1 Rolls. - Each roll of fuse shall be marked to show the contractor's name, lot number, inspector's initials, and the date of manufacture.

5.3.2 Shipping containers. - In addition to any special marking required by the contract or order, and the ICC regulations, shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the title, number, and date of this specification.

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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